Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6	*(a)*			**Indicative content**: **STATION A** Station A will detect both seismic waves as they are both able to travel through the solid mantle. The P wave will arrive first, then the S wave. S waves travel at a slower speed than P waves. This results in a small time lag between the detection of the P and S waves. **STATION B** Station B will detect the waves in the same order as station A. However, the P waves will be detected at a later time than station A because station B is further away from the earthquake. Compared to station A, there will be a larger time lag between the detection of the P and S waves. This is due to the greater distance travelled. **STATION C** Station C will detect P waves but no S waves. It will be the last station to detect the P waves as it is furthest from the origin of the earthquake. P waves are able to travel through both liquid and solid rock. It will not detect any S waves as they are unable to travel through the liquid rock in the Earth's core. The amplitude of the detected waves will decrease as the distance from the earthquake increases. **5-6 marks** Detailed description/explanation of both seismic waves at all three stations. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.* **3-4 marks** A description/explanation of both of the seismic waves detected at two stations **OR** a limited description of both seismic waves at all three stations. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.* **1-2 marks**	3	3		6		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				A limited description/explanation of the seismic waves detected at one / two stations **OR** some description of properties only. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** *No attempt made or no response worthy of credit.*						
	(b)			Vibrations or oscillations (1) In a transverse wave these are 90° to [the direction] the wave [travels] but in a longitudinal wave they are parallel / same direction (1)	2			2		
				Question 6 total	**5**	**3**	**0**	**8**	**0**	**0**